

WILLIS ISLAND, Australia

WMO: 942990

Lat: 16.288S

Lon: 149.965E

Elev: 10

StdP: 101.21

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.4	21.9	11.9	8.7	22.6	13.1	9.4	23.1	14.4	23.0	13.8	23.4	8.4	160	0.755

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.4	31.5	26.4	31.0	26.2	30.5	25.9	27.5	30.0	27.1	29.7	26.8	29.4	5.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.4	29.1	26.3	21.8	28.9	26.0	21.4	28.8	87.4	30.3	85.7	29.9	84.3	29.5	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	13.1	12.3	20.3	32.4	0.7	0.7	19.8	32.8	19.4	33.2	19.1	33.6	18.6	34.1		
(5)			14.8	28.2	0.8	0.4	14.2	28.5	13.8	28.7	13.3	28.9	12.8	29.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.5	28.5	28.5	27.9	27.1	26.0	24.9	24.3	24.2	25.1	26.2	27.3	28.3	(6)	
	DBStd	1.80	0.91	0.94	1.00	0.82	0.86	1.01	0.99	0.91	0.86	0.86	0.79	0.94	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6023	572	517	554	513	497	447	442	440	452	503	519	567	(10)	
	CDD18.3	2982	314	284	296	263	238	197	184	182	202	244	269	309	(11)	
	CDH23.3	26071	3671	3341	3281	2594	1860	1056	681	610	1035	1874	2596	3473	(12)	
(13)	CDH26.7	5747	1233	1141	905	424	121	23	6	7	46	231	528	1083	(13)	
(14)	Wind	WSAvg	8.0	6.7	6.3	7.7	9.0	9.1	9.2	8.8	8.1	7.9	7.4	6.7	(14)	
(15) Precipitation	PrecAvg	1019	189	201	188	111	74	49	33	15	11	15	37	102	(15)	
	PrecMax	1669	637	549	582	392	359	197	222	75	65	87	355	360	(16)	
	PrecMin	241	13	10	7	11	0	0	0	0	0	0	1	0	(17)	
	PrecStd	328	154	148	141	85	78	47	36	16	14	20	58	96	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.1	31.4	29.9	29.0	28.1	27.3	27.5	28.4	29.7	30.9	32.0	(19)	
		MCWB	26.7	26.7	26.6	26.0	24.8	24.1	23.1	23.0	23.9	24.9	25.7	26.5	(20)	
	2%	DB	31.4	31.2	30.7	29.4	28.2	27.3	26.7	26.7	27.7	29.1	30.1	31.2	(21)	
		MCWB	26.4	26.4	26.4	25.5	24.1	23.2	22.4	22.2	23.0	24.4	25.0	26.0	(22)	
	5%	DB	30.8	30.7	30.1	29.0	27.8	26.8	26.2	26.2	27.2	28.6	29.6	30.7	(23)	
		MCWB	26.1	26.3	26.1	25.1	23.7	22.7	22.0	21.6	22.4	23.9	24.6	25.7	(24)	
	10%	DB	30.2	30.3	29.5	28.5	27.4	26.3	25.7	25.7	26.7	28.0	29.1	30.2	(25)	
		MCWB	25.7	26.1	25.8	24.8	23.4	22.4	21.6	21.2	22.0	23.3	24.2	25.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	28.0	27.7	28.2	26.0	25.3	24.6	25.3	25.2	25.9	27.2	27.5	(27)	
		MCDB	30.6	30.7	30.2	28.6	27.7	27.0	26.2	25.8	27.1	28.6	29.0	30.6	(28)	
	2%	WB	27.2	27.4	27.1	26.7	25.2	24.4	23.7	24.0	24.2	25.3	26.2	26.9	(29)	
		MCDB	30.0	30.2	29.7	28.2	27.3	26.4	25.8	25.7	26.6	27.8	28.7	30.0	(30)	
	5%	WB	26.9	27.0	26.7	26.1	24.7	23.8	23.1	23.1	23.6	24.8	25.7	26.5	(31)	
		MCDB	29.7	29.8	29.2	28.0	27.0	26.0	25.5	25.3	26.3	27.4	28.3	29.5	(32)	
	10%	WB	26.6	26.7	26.3	25.6	24.2	23.4	22.6	22.3	23.0	24.3	25.2	26.2	(33)	
		MCDB	29.3	29.4	28.7	27.8	26.7	25.7	25.2	24.9	25.9	27.1	27.9	29.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	3.4	3.1	2.8	2.6	2.6	2.7	3.0	3.4	3.6	3.7	3.8	(35)	
		MCDBR	4.1	3.8	3.5	3.0	2.8	2.8	2.9	3.3	3.7	3.9	4.0	4.1	(36)	
	5% WB	MCWBR	2.0	2.0	2.0	2.0	1.9	1.9	1.9	2.1	2.0	1.9	1.9	1.9	(37)	
		MCWBR	3.7	3.6	3.3	3.0	2.6	2.7	2.7	3.0	3.5	3.7	3.9	3.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.387	0.389	0.395	0.390	0.377	0.378	0.360	0.364	0.373	0.388	0.394	0.386	(40)	
		taud	2.614	2.623	2.590	2.583	2.583	2.574	2.597	2.578	2.533	2.510	2.524	2.587	(41)	
	Ebn at Noon		957	946	919	889	867	847	875	901	925	936	944	957	(42)	
		Edh at Noon	103	101	101	96	90	88	88	95	106	112	112	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.55	6.37	5.68	5.28	4.63	4.15	4.49	5.35	6.38	7.00	7.20	7.14	(44)	
	RadStd		0.61	0.59	0.45	0.39	0.34	0.31	0.25	0.34	0.21	0.39	0.54	0.61	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page